LOCAL APPROACH TO ENTROPY PRODUCTION

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Studienstiftung des deutschen Volkes

Motivation: ¿entropy production rate?

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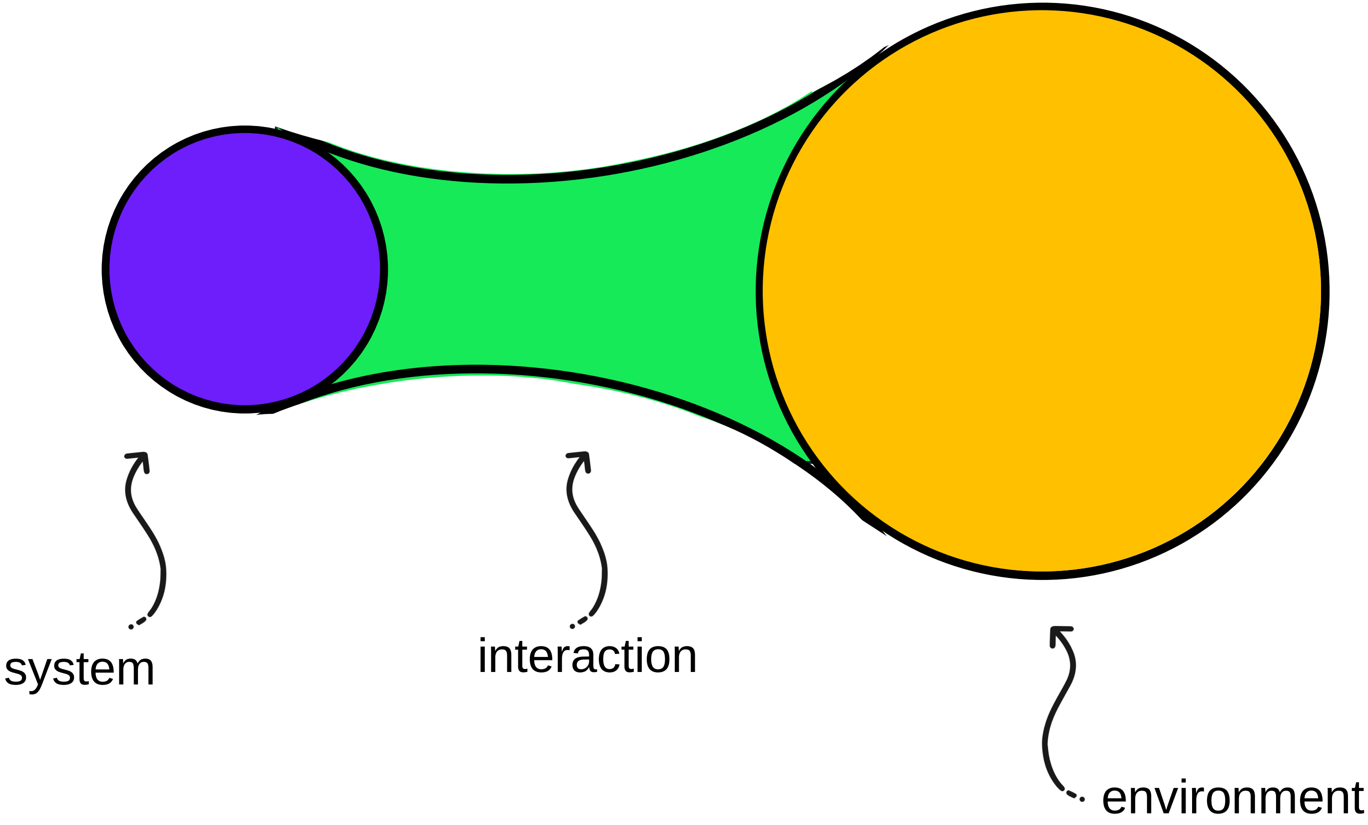
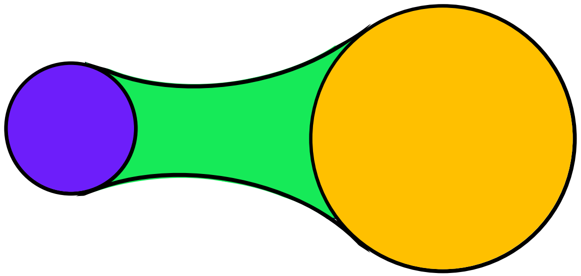
→ *we have to make a choice in the splitting between these two quantities*

Open Quantum Systems

$$H = H_S + H_I + H_E$$



Theory of open quantum systems



Global evolution:

$$\dot{\rho}_{SE}(t) = -i\left[H, \rho_{SE}(t)\right]$$

Initially uncorrelated state

$$\rho_{SE}(0) = \rho_S(0) \otimes \rho_E(0)$$

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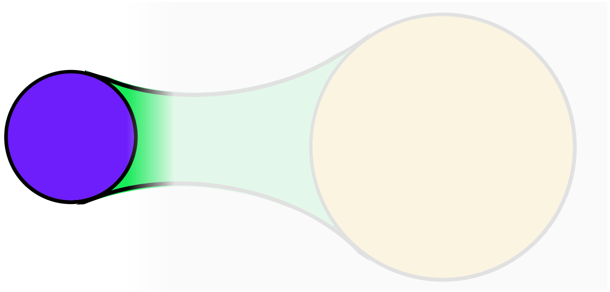

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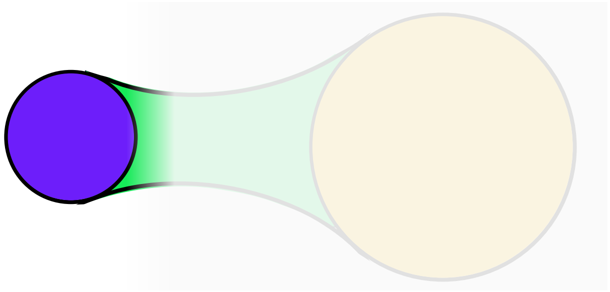

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→ Propagator: $\Phi_{t,s} = \Phi_t \Phi_s^{-1}$

- map is CP-div
- map is P-div ≡ Markovian

S. Wißmann, H.-P. Breuer, and B. Vacchini, Phys. Rev. A 92, 042108 (2015).

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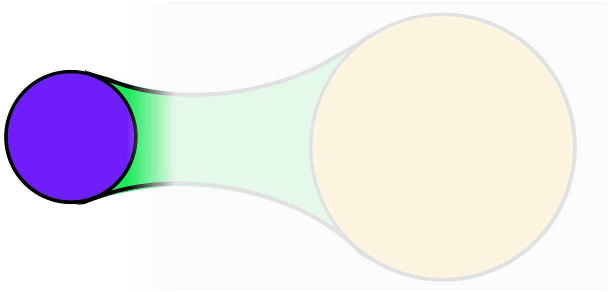

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Time-convolutionless (TCL) master equation

generator

$$\frac{d}{dt}\rho_S(t) = \dot{\Phi}_t\Phi_t^{-1}\left[\rho_S(t)\right] \equiv \mathcal{L}_t\left[\rho_S(t)\right]$$

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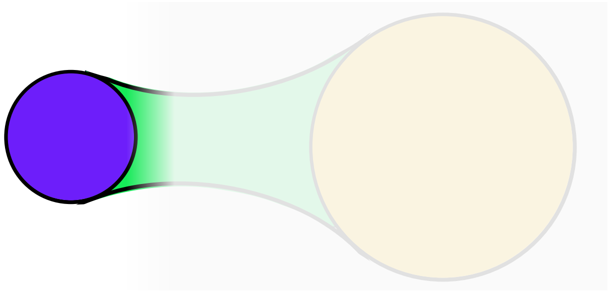
Spectral decomposition of the generator



$$\mathcal{L}_t[\sigma_i(t)] = \underbrace{\lambda_i(t)}_{\text{eigenvalue}}\underbrace{\sigma_i(t)}_{\text{eigenmatrix}}$$

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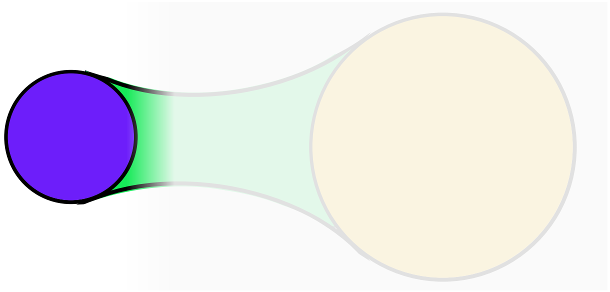

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The generator is:

- Hermiticity preserving $\Rightarrow \mathcal{L}_t[\sigma_i^\dagger(t)] = \lambda_i^*(t)\sigma_i^\dagger(t)$

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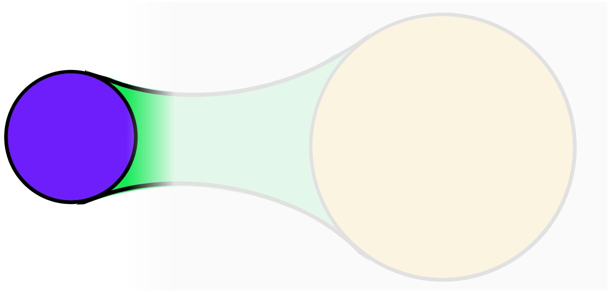

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The generator is:

- Hermiticity preserving $\Rightarrow \mathcal{L}_t[\sigma_i^\dagger(t)] = \lambda_i^*(t)\sigma_i^\dagger(t)$
- Trace destroying $\Rightarrow \mathcal{L}_t^\dagger[\mathrm{Id}] = 0$

$$\Rightarrow \exists\, \rho_S^\star(t) \text{ such that } \mathcal{L}_t[\rho_S^\star(t)] = 0$$

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Weak coupling entropy production rate: Spohn formulation

- CP semigroups: $\Phi_{t+s} = \Phi_t \Phi_s$ → the generator and the fixed point are time-independent

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$$\boxed{\sigma_S(t) = -\left.\frac{d}{d\tau}\right|_{\tau=0} D\left(\Phi_\tau[\rho_S(t)]||\rho_S^\star\right)}$$

relative entropy

$$D(\rho||\sigma) = \mathrm{Tr}\{\rho(\log\rho - \log\sigma)\}$$

fixed point

$$\mathcal{L}[\rho_S^\star] = 0$$

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$$= -\left.\frac{d}{d\tau}\right|_{\tau=0} D\left(\Phi_\tau[\rho_S(t)]||\Phi_\tau[\rho_S^\star]\right) \underline{\geq 0}$$

relative entropy is contractive under CP maps

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Proposal: local entropy production rate for arbitrary scenarios

Instantaneous fixed point (IFP) $\mathcal{L}_t[\rho_S^\star(t)] = 0$

I. A. Picatoste, A. Colla, and H.-P. Breuer, "Local approach to entropy production in the nonequilibrium dynamics of open quantum systems," (2026), arXiv:2603.01861 [quant-ph]

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Assumption: IFP is a state

→ Define the instantaneous map $\Phi_\tau^{(t)} \equiv e^{\tau \mathcal{L}_t}$

$$\sigma_S(t) = -\left.\frac{d}{d\tau}\right|_{\tau=0} D\left(\Phi_\tau^{(t)}[\rho_S(t)]||\rho_S^\star(t)\right)$$

$$= -\left.\frac{d}{d\tau}\right|_{\tau=0} D\left(\rho_S(t+\tau)||\rho_S^\star(t)\right)$$

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$$= \mathrm{Tr}\{\mathcal{L}_t[\rho_S(t)][\log\rho_S(t) - \log\rho_S^\star(t)]\}$$

$$= \dot{S}_S(t) + \underbrace{\mathrm{Tr}\{\dot{\rho}_S(t)\log\rho_S^\star(t)\}}_{\phi_S(t)}$$

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¿Relation to Clausius?

- When the IFP can be written as $$\rho_S^\star(t) = \frac{e^{-\beta H}}{Z}$$

  with a Hamiltonian such that $$\dot{Q}_S(t) = \mathrm{Tr}\{H\dot{\rho}_S(t)\}$$

$$\Rightarrow \sigma_S(t) = \dot{S}_S(t) - \underbrace{\beta\dot{Q}_S(t)}_{\phi_S(t)}$$

Connection between first and second law quantities!

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Connection between first and second law quantities!

¿Why not directly use Clausius then?

- No general definition of temperature.
- Second law independent of the first.

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¿What is the second law?

In this work: $\sigma_S(t) \geq 0 \; \forall \; \rho_S(0)$

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It can be shown:

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upholding of the second law implies negative real part of the eigenvalues of the generator

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$$\text{P-div} \Rightarrow \Phi_\tau^{(t)} \text{ is positive}$$

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$¿ \Leftarrow ?$ Counterexample

Phase-covariant model

$$\mathcal{L}_t[\rho_S(t)] = -i\left[\frac{\omega_r(t)}{2}, \rho_S(t)\right]$$
$$+\gamma_+(t)\left[\sigma_+\rho_S(t)\sigma_- - \frac{1}{2}\{\sigma_-\sigma_+, \rho_S(t)\}\right]$$
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non-P-div

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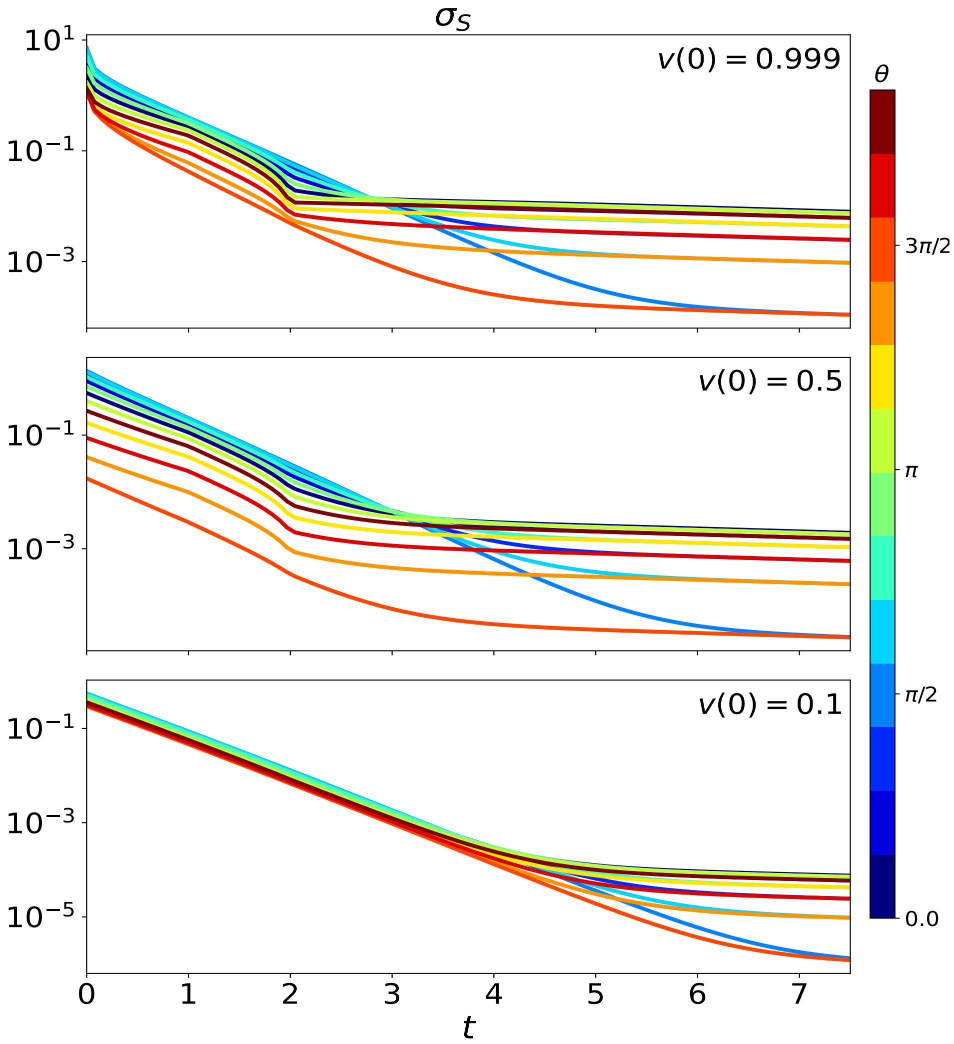

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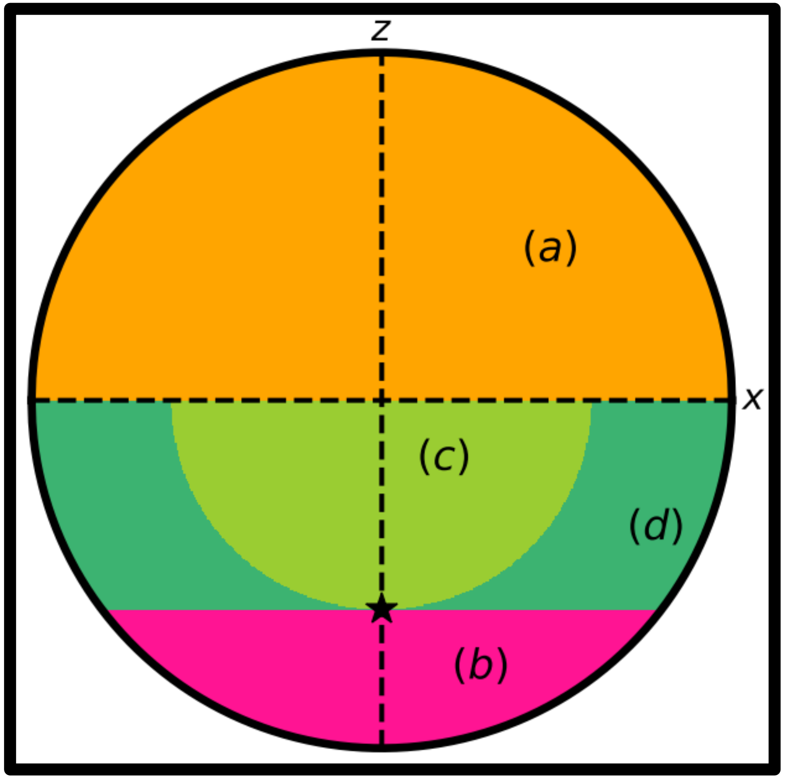

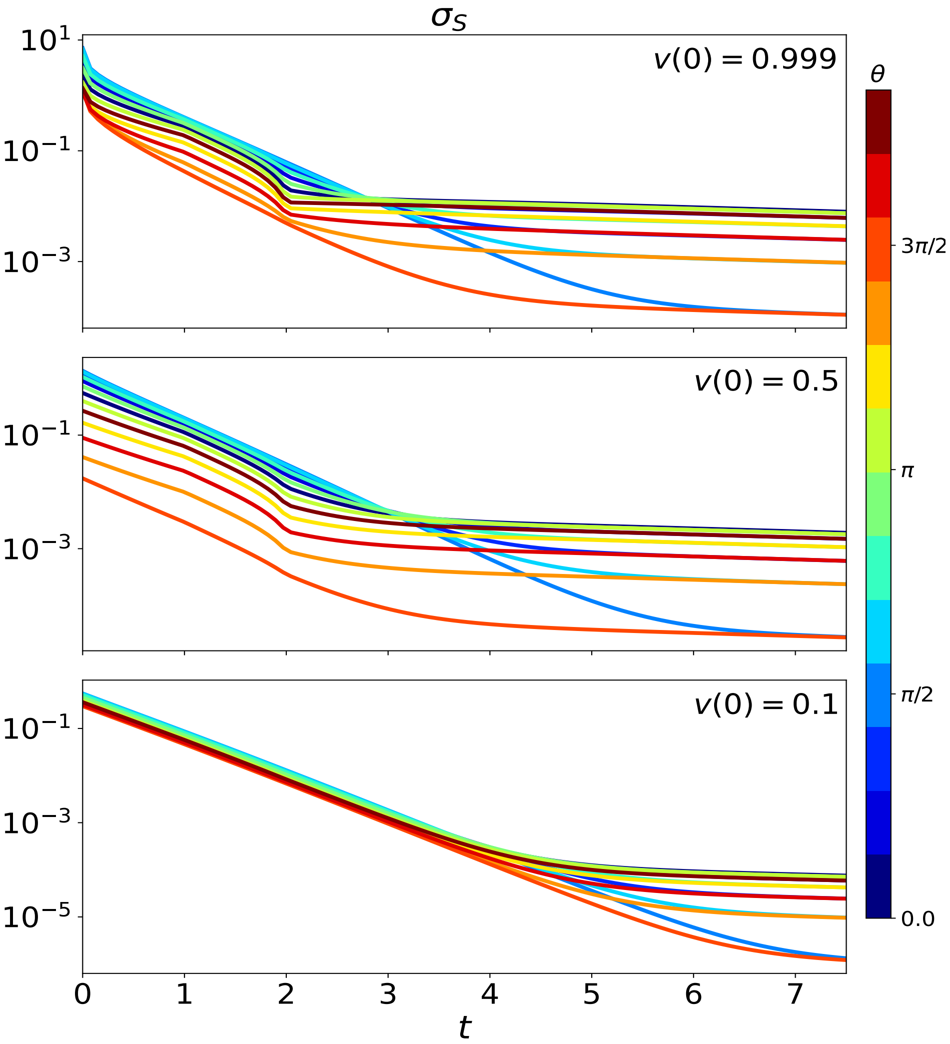

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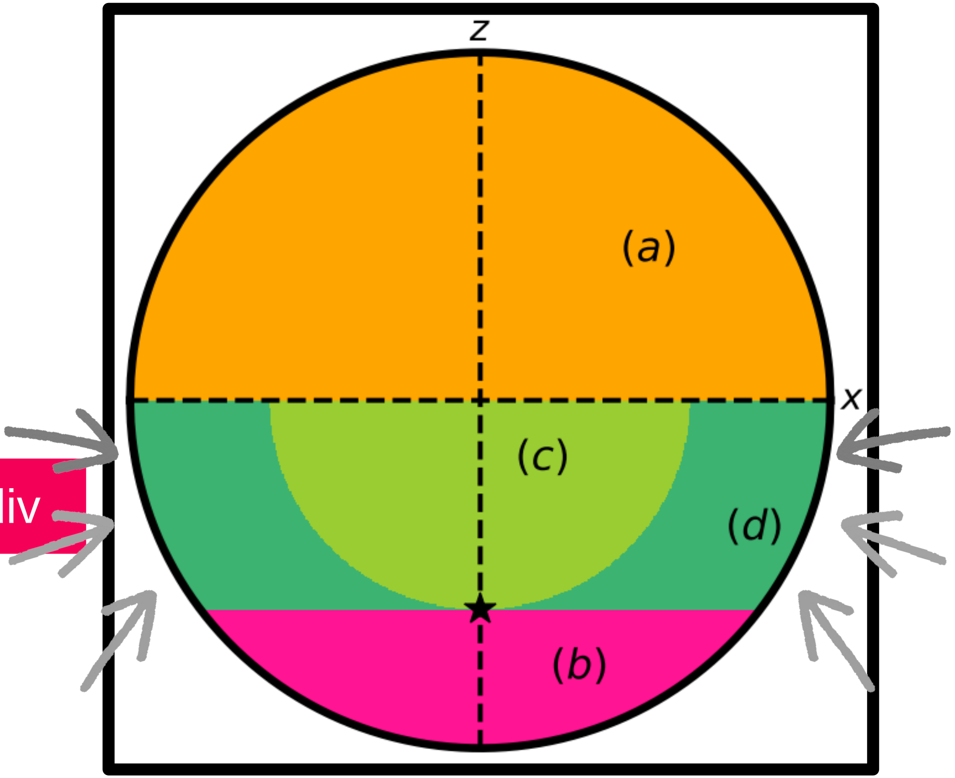

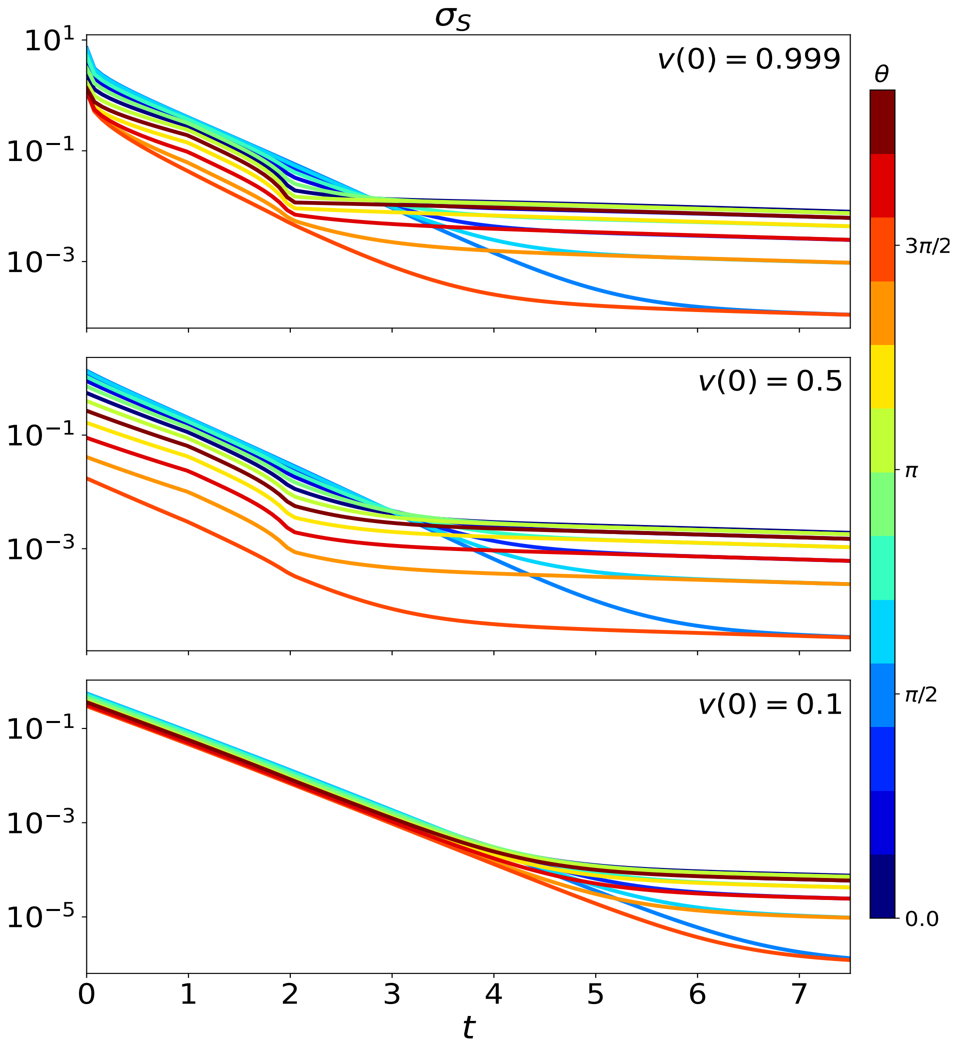

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Map entropy production

$$\sigma_{\text{map}}(t) = \min_{\rho_A} \{-\text{Tr}\{\mathcal{L}_t[\rho_A][\log \rho_A - \log \rho_S^\star(t)]\}\}$$

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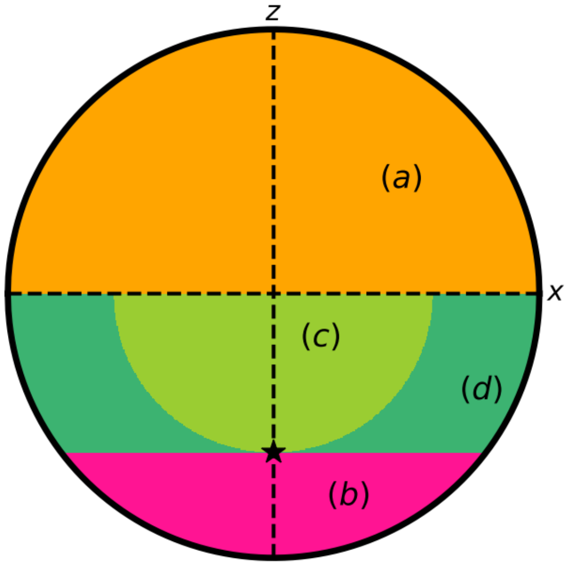

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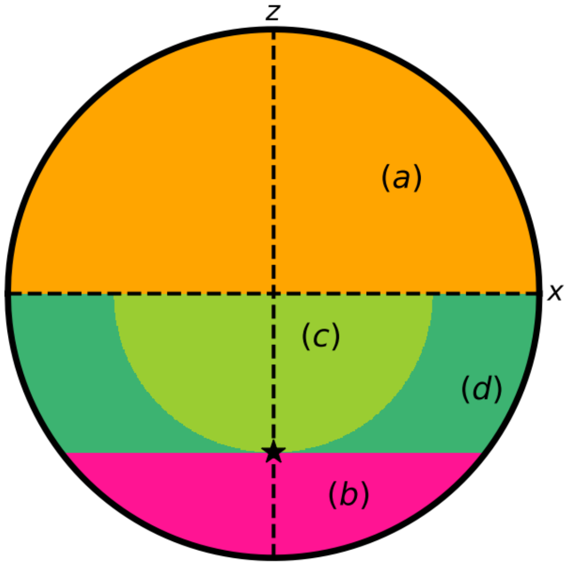
- P-div $\Rightarrow \sigma_{\mathrm{map}}(t) \geq 0$
- non-P-div $\Rightarrow \sigma_{\mathrm{map}}(t) < 0$

$$\iff \langle m|\mathcal{L}_t[|n\rangle\langle n|]|m\rangle < 0$$



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Conclusions

- Local definition of entropy production rate
- Connection between the second law, memory effects and and the spectrum of the generator
- Map entropy production, equivalent to P-div